

pValue and p2Value Storyworlds

Relational Manifolds for Multi-Agent Negotiation Forecasting

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This note positions pValue/p2Value storyworld models as a lightweight relational state representation for coalition, betrayal, and threat forecasting in adversarial multi-agent environments.

Abstract

This note describes a compact representation for social belief dynamics in negotiation storyworlds. The core state separates stable agent traits from directional first-order beliefs (*pValues*) and second-order meta-beliefs (*p2Values*). The resulting representation supports event-driven updates, manifold scans for structural vulnerability, and auditable action traces. The framing is intended as a bridge between storyworld-native scripting, recursive multi-agent social reasoning, and broader Diplomacy-style evaluation work.

1. Summary

The central proposal is to model:

- **Base traits:** stable agent-level coordinates such as loyalty, risk tolerance, and promise keeping.
- **pValues:** direct, directional beliefs such as “A’s estimate of B’s loyalty.”
- **p2Values:** second-order beliefs such as “A’s belief about B’s belief regarding C.”

These quantities are propagated through a simulation loop that performs a **manifold scan** (weakness inference, asymmetric vulnerability, triangle conflicts, and threat ranking), evaluates a small action set with a decomposed utility model, and emits **auditable traces** and **aggregate metrics**. The package aligns directly with the bounded social-reasoning work already collected in the local `GPTStoryworld/social-reasoning` folder, especially the pack overview and recursive-reasoning lane [1, 2].

2. Related Work and Local Positioning

This note sits at the intersection of three nearby lines of work.

First, Diplomacy and no-press strategy papers establish the benchmark pressure: long-horizon coalition management, private belief modeling, and adversarial planning under partial observability [10, 11, 12, 17, 19]. Second, multi-agent LLM frameworks and surveys provide broader context for emergent social behavior, memory, and distributed reasoning [13, 14, 15, 16, 18]. Third, the local storyworld papers and notes extend those concerns into Storyworld-native evaluation, SAE instrumentation, and decoder-oriented forecast tooling [4, 5, 6, 3, 1, 2].

The specific contribution here is narrower than a general MAS framework: it proposes a state representation that is small enough to wire directly into storyworld desirability/effect scripts, but expressive enough to capture betrayal shocks, meta-belief drift, and coalition geometry.

Just as importantly, the local scripts show that this is no longer only a conceptual layer. The existing toolchain already points toward four executable lanes: reusable storyworld template generation, local snapshot scoring, N-agent regime sweeps, and streamed recursive decision traces [7, 8, 9, 2]. Those scripts imply that future work should focus less on inventing new ontology and more on tightening the loop between relational state, storyworld authoring, forecast metrics, and trace-grounded evaluation.

3. Representation: Base Traits + Relational Fibers

3.1 Base personality coordinates

Each agent i has a latent trait vector

$$\theta_i \in \mathbb{R}^d.$$

Example trait axes include loyalty, risk tolerance, reciprocity, aggressiveness, and promise keeping.

3.2 First-order relational beliefs (pValues)

For each ordered pair (observer = i , target = j) and trait k ,

$$p[i, j, k] \in [0, 1].$$

Interpretation: $p[A, B, \text{loyalty}]$ is A's estimate of B's loyalty.

3.3 Second-order meta-beliefs (p2Values)

For each ordered triple (observer = i , model = j , target = m) and trait k ,

$$p2[i, j, m, k] \in [0, 1].$$

Interpretation: $p2[A, B, C, \text{loyalty}]$ is A's belief about B's belief regarding C's loyalty.

3.4 Product manifold view

A compact view of the full state is

$$\mathcal{M} = \left(\prod_i \theta_i \right) \times \left(\prod_{i \neq j} p[i, j, :] \right) \times \left(\prod_{i \neq j \neq m} p2[i, j, m, :] \right).$$

This factorization makes the modeling choice explicit: base traits occupy global coordinates, while p and $p2$ behave like lightweight relational fibers over that base state. That matches the local social-reasoning pack's emphasis on simple state objects that can still support coalition, betrayal, and death-ground analysis [1].

4. Dynamics: Event-Driven Updates

4.1 Direct pValue update

When an event occurs (actor u takes action a), an observer i updates its belief about u via

$$p_{t+1}[i, u, k] \leftarrow \text{clip}\left((1 - \alpha_k) p_t[i, u, k] + \alpha_k \text{signal}_k(e_t, i), 0, 1\right).$$

Here signal_k maps an observed action into trait-specific evidence.

4.2 p2Value update

Second-order beliefs drift toward an inferred model of another agent’s direct beliefs:

$$p_{2t+1}[i, j, m, k] \leftarrow \text{clip}\left((1 - \beta_k) p_{2t}[i, j, m, k] + \beta_k \widehat{p[j, m, k]}, 0, 1\right).$$

4.3 Surprise weighting and trusted betrayal

In negotiation settings, updates should not be symmetric. A sharp negative action can trigger an amplified belief revision when prior trust was high. In the recursive-reasoning lane, this is operationalized through surprise amplification, trusted-betrayal collapse, and alliance-constraint penalties [2]. This asymmetry is one of the main reasons to prefer relational state variables over static trait-only models.

5. Decision Layer: Manifold Scan $\rightarrow$ Utility $\rightarrow$ Action

5.1 Manifold scan

Before choosing an action, the agent scans for structural risks:

- **Weakness inference:** who appears to think I am weak?
- **Asymmetric vulnerability:** I trust them more than I believe they trust me.
- **Triangle conflicts:** I trust both B and C, but B and C are in high conflict.
- **Threat ranking:** which opponent is most likely to exploit me?

These scan outputs are logged in the trace and can be reused as selection features. This makes the pValue/p2Value layer compatible with both explicit rule-based selectors and learned controllers.

5.2 Utility decomposition

For each candidate action, the model evaluates a decomposed utility score (gain, risk, reputation, coalition bonuses, aggression bonuses, and so on) and then selects an action. The local recursive package also permits epsilon exploration so that non-top choices occasionally appear among the best few candidates, yielding less brittle behavior and richer traces [2].

6. Storyworld Integration: Keyrings

To make $p/p2$ values storyworld-native, properties can be referenced via stable keyrings:

- **pValue keyring:** [property_id, perceived_character_id]
- **p2Value keyring:** [property_id, perceived_character_id, target_character_id]

This lets desirability scripts and after-effects read and write relational beliefs without bespoke plumbing. It also lines up with the local rule- and decoder-oriented Storyworld work, which treats storyworld state as something both forecastable and inspectable rather than hidden behind opaque agent context [3, 5].

7. Worked Micro-Example from an Actual Trace

Setup: example storyworld `diplomacy_min.json` with agents P1, P2, and P3. A 6-turn recursive MAS simulation was run, and turns 1–3 were extracted for agent P2.

Turn 1 (P2 acts)

- **Scan (high-level):** `weak_from_max` $\approx$ 0.87, top threat = P3
- **Top-3 candidates:**
 1. `propose_coalition(P1)` (0.133)
 2. `commit_total_war()` (0.069)
 3. `betray(P1)` (0.049)
- **Chosen action:** `betray(P1)`.
- **Reason non-top action fired:** epsilon exploration.

Turn 2 (P2 acts)

- **Scan:** `weak_from_max` $\approx$ 0.88, top threat = P3
- **Top-3 candidates:** `propose_coalition(P1)` > `commit_total_war()` > `betray(P3)`

- **Chosen action:** propose_coalition(P1).
- **Interpretation:** after the exploratory betrayal, the model attempts to restabilize via coalition-building against the top threat.

Turn 3 (P2 acts)

- **Scan:** weak_from_max $\approx$ 0.91, top threat = P3
- **Top-3 candidates:** commit_total_war() > propose_coalition(P1) > betray(P1)
- **Chosen action:** commit_total_war().
- **Interpretation:** rising perceived weakness and threat pushes the posture toward hard escalation.

Raw trace excerpt (abridged)

```
{
  "turn": 1,
  "agent": "P2",
  "action": "betray",
  "target": "P1",
  "decision": {
    "scan": {
      "weak_from": {
        "P1": 0.338,
        "P3": 0.871
      },
      "threat_rank": [
        ["P3", 0.733],
        ["P1", 0.331]
      ]
    },
    "candidates_top3": [
      {
        "action": "propose_coalition",
        "target": "P1",
        "score": 0.13308
      },
      {
        "action": "commit_total_war",
        "target": null,
        "score": 0.068644
      }
    ]
  }
}
```

```

        "action": "betray",
        "target": "P1",
        "score": 0.048735
    }
],
"rationale_text": "weak_from_max=0.87; triangle_conflicts=0; snap=0.00; top_threat=P3; cho
}
}

```

8. What to Add Next

1. Calibration and histograms

- Confidence histograms for coalition acceptance and betrayal likelihood.
- Reliability curves for a small binary outcome set.

2. Outcome-centric plots

- Instability index vs. meta-consistency scatter.
- Death-ground entries vs. asymmetric vulnerability count.

3. Selector experiments

- Replace the fixed utility action chooser with a storyworld-conditioned selector:

$$S_t = \arg \min_{S \in \mathcal{S}} \mathbb{E} [\text{Brier}(p(S, \phi_t), y)].$$

- Measure whether form-fit improves Brier under distribution shift across storyworld families.

4. Compiled storyworld authoring

- Promote the current template generator into a true compiler from relational motifs to storyworld fragments. The existing script already emits desirability and after-effect blocks for coalition, defection, and betrayal; the next step is to generate larger encounter bundles, not just local fragments [7].
- Treat pValue and p2Value keyrings as an authoring IR rather than an implementation detail. That would allow one layer to specify social geometry and a downstream layer to emit concrete storyworld JSON.

5. Fast forecast probes

- The snapshot scoring script already computes coalition, defection, and betrayal probabilities plus a global strategic-tension score from a single $p/p2$ state [8]. A natural next move is to use those scores as cheap proposal filters before invoking full recursive reasoning.
- This suggests a two-stage architecture: first score local manifold states quickly, then spend expensive simulation or LLM budget only on the most ambiguous or unstable regions.

6. Regime maps instead of single trajectories

- The N-agent simulation series is already set up to sweep 4–7 player settings while tracking instability, cohesion, Paine violations, death-ground entries, burn-the-boats signals, and asymmetric vulnerability events [9]. The stronger future-work version is to treat those outputs as a phase diagram rather than a bag of episodes.
- In other words: map where coalition order, betrayal cascades, and death-ground dynamics dominate as a function of agent count, surprise sensitivity, and survival pressure. That would make the manifold view empirically legible.

7. Trace-grounded social interpretability

- The recursive reasoner logs explicit decision traces, rationale summaries, top candidates, and threat scans without relying on hidden chain-of-thought [2]. That is a stronger starting point for interpretability than most black-box MAS work.
- A promising direction is to align SAE features or other probes against those explicit trace fields: weak-score spikes, triangle conflicts, derivative snap, death-ground mode, and action flips. This would connect relational state, trace semantics, and internal model instrumentation.

8. Internal Storyworld links

- Tie the relational manifold more directly to the SAE instrumentation and decoder/evaluation framing in the local Storyworld papers [4, 5, 6].
- Connect the manifold state to explicit rule bundles and constitutional or verifier-facing axes rather than leaving it as a purely latent forecast object [3].

A. Operational Appendix: Focused Diplomacy Harness

The same relational-state framing also suggests a practical evaluation doctrine for Diplomacy-style runs: do not benchmark the entire game when the scientific question is local social forecasting. In the first focused February harness passes, the useful move was to pin a strategically dense 1915 press scenario, restrict the focus set to a few powers, keep a single negotiation round, and evaluate only the movement phases that actually carried the coalition and betrayal decisions. The result was a materially smaller call surface than the earlier broad runs, without deleting the negotiation logic that mattered.

This matters because short storyworlds were not functioning as full alternative games. They behaved more like compact rhetorical forecast objects: coalition warnings, false-concession traps, and backstab priors that could be injected into message generation and diary interpretation. The key benchmark question therefore becomes not merely whether a storyworld was assigned, but whether it became explicit negotiation content. In the strongest focused runs, fully explicit uptake aligned with lower forecast error; when uptake fell back toward implicit context, forecast quality degraded.

That operational lesson fits the general thesis of this note. pValue and p2Value structures are useful not only because they provide a cleaner ontology for social state, but because they make it easier to

stage narrow, auditable experiments where social geometry is held in focus and narrative scaffolds can be evaluated as forecast instruments rather than decorative prompting.

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